

MATH 130: 10/1 WORKSHEET

The past few weeks we've been learning about truth tables and propositional logic—the logic where we can use and, or, not, if-then, and if-and-only-if to combine propositions. Today we look at the limitations of this logic and what we need to do to get a more expressive language.

- All humans are mortal.
- Socrates is human.
- $\therefore$ Socrates is mortal.
- Every positive number has a square root.
- 2 is positive.
- $\therefore$ 2 has a square root.

The language of predicate logic.

Predicate logic extends propositional logic by having variables $x, y, \dots$ and names for individuals and *prediacte symbols* $P(x), Q(x, y), \dots$ to stand for statements about individuals.

- $M(x)$ means “ x is mortal”.
- $M(s)$ means “Socrates is mortal”.
- $R(x, y)$ means “ x is the square root of y ”.
- $R(2, 4)$ means “2 is the square root of p ”.

Predicate logic also introduces *quantifiers* to make general statements about individuals.

- (Universal quantifier) $\forall x P(x)$ means “every individual x has property P ”.
- (Existential quantifier) $\exists x P(x)$ means “there is at least one individual x with property P ”.

For example, we could express “All humans are mortal” as $\forall x M(x)$.

We can combine quantifiers and connectives to make complicated statements. For example, if $P(x)$ means “ x is positive” and $R(x, y)$ means “ x is the square root of y ”, then we could express “every positive number has a square root” as

$$\forall x P(x) \Rightarrow \exists y R(y, x).$$

Read literally: for every x if x is positive then there is a y so that y is the square root of x .

“Truth tables” for predicate logic?

With propositional logic, we had truth tables as a device to exhaustive list all possibilities for a statement. How could we do the same with predicate logic?

- With propositional logic, a possible meaning of a sentence was the truth value of its propositional variables.
- With predicate logic, a possible meaning needs to say what the domain of individuals is and what the meaning of the predicate symbols is.

Instead of having only a few possible meanings—namely, 2^n possible meanings if there are n many propositional variables—now there are infinitely many possible meanings. So there's too many to make a table to exhaustively list all possibilities.

Predicate logic is more expressive than propositional logic. Indeed, while propositional logic is inadequate for doing mathematics, predicate logic is strong enough. But this extra power comes at a cost. Predicate logic is more difficult to analyze.